

RALEIGH EXECUTIVE, NC, USA

WMO: 722201

Lat: 35.582N

Lon: 79.101W

Elev: 75

StdP: 100.42

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.7	-5.9	-17.2	0.8	-2.3	-13.9	1.1	-0.8	8.9	11.8	8.0	11.2	0.6	320	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	35.7	23.8	34.0	23.6	32.7	23.4	26.1	31.3	25.4	30.6	24.9	29.9	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.1	27.7	24.0	19.1	27.1	23.6	18.6	26.8	80.7	31.5	78.0	30.8	76.1	29.4	36.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.5	6.2	5.3	DB	-12.1	37.3	3.2	1.7	-14.5	38.5	-16.3	39.5	-18.1	40.5	-20.5	41.7
(5)				WB	-11.6	27.3	3.6	0.6	-14.2	27.8	-16.3	28.1	-18.3	28.4	-21.0	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	5.1	6.6	10.3	15.6	20.3	24.7	26.4	25.7	22.6	16.0	9.7	6.9	(6)
(7)		DBStd	8.76	5.64	5.19	5.39	4.51	3.62	2.74	2.25	2.58	3.12	4.74	4.63	5.07	(7)
(8)		HDD10.0	558	172	121	65	7	0	0	0	0	7	61	126		(8)
(9)		HDD18.3	1844	411	330	253	104	24	1	0	5	102	259	355		(9)
(10)		CDD10.0	2704	19	26	74	176	320	440	509	487	378	193	53	29	(10)
(11)		CDD18.3	951	0	1	4	23	86	191	251	229	133	31	2	1	(11)
(12)		CDH23.3	9322	1	7	66	297	831	1947	2562	2174	1148	271	16	3	(12)
(13)		CDH26.7	3926	0	0	7	72	280	870	1211	993	431	63	1	0	(13)
(14)	Wind	WSAvg	1.6	1.8	1.9	2.1	2.1	1.6	1.4	1.3	1.1	1.4	1.5	1.5	(14)	
(15)	Precipitation	PrecAvg	1180	91	80	106	82	84	110	132	123	114	93	80	78	(15)
(16)		PrecMax	1512	198	161	194	165	187	316	217	273	419	237	203	159	(16)
(17)		PrecMin	953	24	26	29	31	38	24	75	27	20	3	13	29	(17)
(18)		PrecStd	156	47	34	46	36	35	69	39	53	94	61	51	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	24.1	27.5	31.1	33.1	36.8	37.4	37.7	34.9	31.3	25.9	23.0	(19)
(20)		MCWB	17.0	17.5	16.3	18.4	21.7	23.7	24.7	25.5	21.5	22.4	17.8	18.2	(20)	
(21)		2%	DB	20.0	21.3	25.2	28.7	31.6	34.2	35.2	35.1	32.7	28.0	22.8	21.0	(21)
(22)		MCWB	15.9	15.1	16.4	18.3	21.1	23.3	24.4	24.3	21.8	20.5	16.5	17.5	(22)	
(23)		5%	DB	17.5	18.8	22.9	27.1	29.9	32.8	33.9	33.2	31.1	26.3	21.1	18.4	(23)
(24)		MCWB	13.6	13.6	14.9	17.6	20.8	23.1	24.3	24.2	22.2	19.8	16.3	16.1	(24)	
(25)		10%	DB	15.0	16.3	20.3	24.7	27.8	31.7	32.6	32.0	29.1	24.1	18.7	16.2	(25)
(26)		MCWB	11.2	11.4	14.3	16.9	20.2	22.7	24.0	24.0	21.9	19.6	15.0	13.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	18.7	19.0	21.9	23.8	26.1	26.9	27.3	25.7	24.1	20.6	20.2	(27)
(28)		MDCB	20.6	22.7	23.8	25.2	28.7	31.5	32.6	32.4	28.7	27.0	22.6	21.6	(28)	
(29)		2%	WB	16.8	16.9	17.9	20.5	22.8	25.1	26.1	26.1	24.9	23.3	18.9	18.3	(29)
(30)		MDCB	19.3	20.0	21.9	24.4	27.7	30.5	31.8	31.1	28.4	25.9	21.0	20.0	(30)	
(31)		5%	WB	14.7	15.2	17.1	19.4	22.0	24.5	25.4	25.5	24.1	22.2	17.3	16.5	(31)
(32)		MDCB	17.0	18.0	20.6	23.8	27.1	30.0	31.1	30.4	27.9	25.1	19.5	18.0	(32)	
(33)		10%	WB	12.1	12.5	15.8	18.3	21.3	23.9	24.9	24.9	23.4	20.6	15.5	14.0	(33)
(34)		MDCB	13.9	15.2	19.6	22.9	26.0	29.3	30.3	29.6	26.8	22.9	18.4	15.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.1	12.7	13.9	14.4	12.8	12.3	11.8	11.1	11.6	13.2	13.9	12.1	(35)	
(36)		5% DB	MCDBR	15.3	15.8	16.9	17.0	15.0	13.9	13.4	13.1	14.3	14.5	16.0	14.1	(36)
(37)		MCWBR	11.0	10.0	8.7	7.5	6.1	4.8	4.6	4.7	5.6	7.8	10.3	10.6	(37)	
(38)		5% WB	MCDBR	13.4	13.9	13.5	12.9	12.3	12.2	11.7	11.3	10.1	10.6	12.4	12.3	(38)
(39)		MCWBR	11.1	10.3	8.3	6.9	5.6	4.8	4.5	4.7	5.0	6.3	9.8	10.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.336	0.370	0.411	0.463	0.506	0.553	0.533	0.432	0.391	0.338	0.320	(40)	
(41)		taud	2.484	2.431	2.370	2.256	2.149	2.070	1.958	1.999	2.279	2.362	2.474	2.516	(41)	
(42)		Ebn at Noon	881	905	898	874	830	791	751	757	824	829	850	858	(42)	
(43)		Edh at Noon	84	99	115	135	153	165	184	173	124	105	84	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.60	3.27	4.32	5.42	6.00	6.39	6.03	5.47	4.59	3.73	2.85	2.25	(44)	
(45)		RadStd	0.17	0.39	0.30	0.39	0.47	0.45	0.37	0.27	0.39	0.49	0.23	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+156	+111

Nomenclature: See separate page